BioMutFed+: Mutation-Driven Federated Learning for IIoT

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***Abstract*—The evolution towards Industry 5.0 underscores the critical need for privacy-preserving and human-centric technologies within Industrial Internet of Things (IIoT) ecosystems. Federated Learning (FL), an essential framework enabling decentralized model training while safeguarding data privacy, continues to encounter significant obstacles such as non-IID data distributions, adversarial threats, and limited scalability. To address these issues, we propose BIOMUTFED+, a biologically inspired federated learning framework featuring adaptive mutation-based gradient perturbations, pheromone-driven adaptive client selection, and robust trimmed-mean aggregation. Comprehensive theoretical analysis guarantees convergence at a rate of $\mathcal{O}(1/\sqrt{T})$. Empirical evaluations on Digits (visual tasks), NASA FD004 (predictive maintenance), and Wisconsin Breast Cancer datasets demonstrate rapid convergence within 20 rounds, variance reductions of 1.2–2.9$\times$ against established FL methods, and enhanced adversarial robustness under 20% malicious client scenarios compared to current approaches. Furthermore, we introduce FEDMUTADAM, a lightweight BIOMUTFED+ variant integrating adaptive gradient clipping with server side Adam style updates, optimized for resource-constrained IIoT devices. Our framework thus represents a scalable and robust solution advancing secure and adaptive intelligent systems for IIoT systems.**

***Index Terms*—Federated learning, Bio-inspired optimization, Privacy-preserving AI, Industry 5.0, IIoT, Mutation-based learning, Adaptive optimization, Adversarial robustness.**

I. INTRODUCTION

ARTIFICIAL Intelligence (AI) is significantly transforming industries when combined with the Internet of Things (IoT). This integration fosters seamless collaboration between human ingenuity and machine intelligence, accelerating the progression of the industrial revolution. Embodied by the vision of Industry 5.0, this paradigm shift emphasizes human-centric collaboration rather than mere automation, addressing complex challenges inherent in Industrial Internet of Things (IIoT) ecosystems [1].

Federated Learning (FL) is a crucial framework for privacy-preserving, distributed intelligence, facilitating collaborative model training across decentralized devices. However, FL encounters substantial challenges in IIoT environments. These challenges include non-Independent and Identically Distributed (non-IID) data, adversarial attacks, and communication overhead. Thus, there is a need for innovative approaches specifically designed for Industry 5.0 [2]–[4].

The evolution of FL has led to Federated X Learning, where "X" represents distinct sectors such as the IoT, healthcare, and finance, each facing unique challenges. Federated X Learning capitalizes on the advantages of FL, including privacy preservation, scalability, and collaborative training, while incorporating methodologies specific to each domain. This specialized approach addresses the unique demands of different industries, promoting innovations that improve decentralized learning and transform industry practices [5]. FL in intelligent IoT has been extensively studied, demonstrating notable advancements in privacy preservation, managing data heterogeneity, and enhancing robustness against adversarial threats [6]. However, existing approaches face critical limitations when deployed in real-world industrial scenarios characterized by extreme non-IID data distributions, where statistical heterogeneity can degrade convergence by up to 40% compared to IID settings [7]. Furthermore, traditional FL aggregation methods demonstrate vulnerability to Byzantine attacks and model poisoning, with performance degradation exceeding 60% under adversarial conditions involving even modest fractions of malicious clients [8], [9].

Recent studies emphasize adaptive FL frameworks that ensure scalability and reliability in dynamic industrial environments [10], [11]. Yet, current solutions struggle with the computational and communication overhead required for large-scale IIoT deployments, where networks may comprise thousands of heterogeneous edge devices with varying computational capabilities and intermittent connectivity [12]. The challenge is further amplified by the need for real-time responsiveness in critical industrial applications, where traditional federated approaches often fail to meet latency requirements due to synchronous aggregation protocols and inefficient client selection mechanisms [13], [14].

To address the multifaceted challenges of FL in IIoT systems particularly under non-IID data, adversarial threats, and resource constraints, we introduce a two-part optimization framework composed of **BIOMUTFED+** and **FEDMUTADAM**. BIOMUTFED+ is a biologically inspired algorithm that introduces server side mutation-based gradient perturbations to enhance generalization and adversarial robustness without increasing communication rounds.

Complementarily, FEDMUTADAM is a distilled variant designed for resource-constrained IIoT nodes, incorporating

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adaptive moment-based updates (inspired by Adam) and norm-clipped mutations for stable and efficient learning. This dual approach effectively addresses the need for robust optimization and lightweight deployment, providing a secure and adaptable federated learning framework that meets the essential demands of Industry 5.0.

Fig. 1 illustrates the architecture of BIOMUTFED+, depicting global model distribution, local training with adaptive mutation perturbations, pheromone-based client selection dynamically updated through reinforcement signals derived from client performance, and robust aggregation mechanisms. This framework enables efficient and secure training among heterogeneous clients, while maintaining resilience against adversarial threats.

The contributions of this article include the following.

1) A Dual Component FL Framework: We propose a novel federated learning framework comprising two complementary algorithms, BIOMUTFED+ for robust optimization and FEDMUTADAM for resource-efficient deployment; targeted at the needs of Industry 5.0 IIoT systems.
2) Theoretical convergence and robustness guarantees: We provide a rigorous theoretical analysis proving that BIOMUTFED+ achieves convergence to a stationary point at a rate of $O(1/\sqrt{T})$ under standard assumptions of smoothness, bounded variance, and controlled client heterogeneity. Additionally, we derive an optimal trimming rate to balance adversarial impact mitigation and information retention, ensuring robust aggregation even under high adversarial fractions.
3) Comprehensive Empirical Validation Across Diverse IIoT Datasets: We extensively validate BIOMUTFED+ on challenging benchmarks representative of real world IIoT applications, including multiple vision tasks (Digits, MNIST, Fashion-MNIST), NASA C-MAPSS FD004 for predictive maintenance, and Wisconsin Breast Cancer dataset for healthcare diagnostics. Our framework consistently outperforms established federated learning baselines with notable variance reductions ($1.2\times$ to $2.9\times$) and improved convergence characteristics under non-IID conditions.
4) Practical Scalability and Energy Efficiency: The pheromone-based client selection mechanism enables efficient and scalable federated training by dynamically prioritizing high-quality clients, reducing communication and computation overhead. Mutation-based perturbations prevent premature convergence, promoting robustness without excessive resource consumption, making BIOMUTFED+ and its distilled version FEDMUTADAM highly suitable for deployment in large-scale and resource-constrained IIoT networks.

The remainder of this paper is organized as follows. Section II reviews recent advancements in FL and places our contribution within the existing literature. Section III formalizes the system model and defines the problem that underpins our study. Section IV describes the proposed framework and includes a thorough theoretical analysis. Section V outlines the experimental methodology and discusses the empirical findings. Section VI presents the key findings in the conclusion, while Section VII suggests directions for future research. ***Note:*** Supplementary material includes extended proofs, additional experiments, and sensitivity analyses.

II. RELATED WORK

This section reviews the literature on FL and bio-inspired optimization, focusing on their limitations in addressing statistical heterogeneity and adversarial threats within IIoT applications. For additional information, please refer to the supplementary file related to this work.

A. *Existing FL Frameworks*

FL has emerged as a promising approach for enabling distributed privacy-preserving training across heterogeneous devices [15]. Foundational FL frameworks such as FedAvg [16] and FedProx [17] primarily focus on communication efficiency and mitigating statistical heterogeneity. However, these methods often struggle to maintain stable convergence and robustness in the face of highly non-IID data distributions common in IIoT environments [18], [19].

Recent advances have introduced sophisticated mechanisms to address these challenges. For instance, SCAFFOLD [20] employs control variates to reduce client drift; however, its computational overhead limits scalability. Domain-specific frameworks like FedML [21] and FATE [22] integrate privacy safeguards and modular designs tailored for industrial and cross-organizational applications. Despite their strengths, these frameworks still encounter challenges related to complex parameter tuning requirements and are actively developing unified strategies for adversarial resilience and dynamic client prioritization.

B. *Bio-Inspired Optimization in FL*

The application of bio-inspired optimization strategies in FL has gained significant momentum in recent years. Evolutionary algorithms have been explored to reduce communication costs [23], while Particle Swarm Optimization (PSO) has demonstrated promising results in healthcare, as shown by Victor et al. [24], who integrated PSO with FL to improve stroke prediction while preserving patient privacy. Additionally, FedDual [25] employs a dual-network architecture inspired by human memory systems, mimicking neocortical and hippocampal processing to address statistical heterogeneity through synthetic data generation. This approach achieves faster convergence and improved accuracy.

Other biological paradigms, such as ant colony optimization and evolutionary techniques [26]–[28], have been employed to refine client selection and exploration strategies. However, many of these methods rely on domain-specific tuning, limiting their applicability across heterogeneous IIoT environments [29].

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C. Adversarial Resilience in FL

Adversarial attacks pose a persistent threat to FL frameworks. Byzantine tolerant aggregation methods such as Trimmed Mean and Krum have been proposed to address this, but their efficacy diminishes in environments with high client variability. Liu et al. tackled this challenge by developing communication-efficient digital twin networks that maintain robustness against adversarial perturbations while enabling human-in-the-loop feedback mechanisms [30]. Their work bridges the gap between automated systems and human expertise, a key principle of Industry 5.0. Recent advances have begun to integrate adversarial detection mechanisms, but these solutions often introduce additional computational overhead. Recent surveys highlight adversarial attacks in Federated Learning, focusing on their impact, stealth, and resource constraints. They address low-budget, low-visibility, high-impact threats and review defense strategies, emphasizing persistent challenges in securing FL systems [31]. More recently, FLTrust [32] introduced trust bootstrapping from a server-held root dataset to score client updates via cosine similarity, while FLAME [33] combines cosine-based outlier filtering with adaptive clipping and differential noise injection. Although effective under moderate heterogeneity, these methods face limitations under severe non-IID conditions and complex temporal architectures, as demonstrated in our experiments (Section V). Recent work has further explored hybrid detection and reputation-based defenses for adversarial FL in industrial and 5G settings [34]–[36]. BIOMUTFED+ extends these principles by introducing hybrid aggregation and mutation mechanisms, thereby achieving resilience without imposing significant computational overhead.

D. FL in IIoT Applications

FL has been applied across diverse IoT and IIoT domains [37]. Recent industrial systems include privacy-preserving collaborative inference for hydrocarbon exploration [38]; LATTE, a middleware that incorporates development-chain diversity into training-time estimation with a data-driven runtime selector [39]; a reinforcement-learning scheduler modeling client incentives for green IIoT [40]; RFL-APIA, a fuzzy-rule framework detecting poisoned updates via gradient-variance scoring [41]; and LotusFL, an on-device system jointly optimizing client selection and batch size to counter watermark-induced shortcut learning [42].

E. Addressing Limitations in Existing FL Methods

The preceding analysis identifies three critical gaps in current FL methodologies for IIoT deployment. *First*, existing client selection mechanisms in FedAvg [16] and SCAFFOLD [20] rely on uniform random sampling or coordination protocols with quadratic complexity, creating bottlenecks in large industrial federations; we address this through linear-cost pheromone-based selection that dynamically tracks client contribution quality. *Second*, methods like FedProx [17] and FedDyn [43] impose extra local computation unsuitable for battery-constrained edge devices; we mitigate this with FEDMUTADAM, which transfers adaptive moment estimation and gradient clipping to the server while preserving convergence performance. *Third*, Byzantine-tolerant methods such as Krum [8] demonstrate limited effectiveness under the high client variability and severe non-IID conditions characteristic of industrial deployments; BIOMUTFED+ integrates pheromone-driven client prioritization with hybrid robust aggregation to provide superior adversarial resilience while retaining information from honest participants. The proposed dual framework—BIOMUTFED+ for resource-rich environments and FEDMUTADAM for constrained settings—collectively establishes new capabilities in scalability and robustness for Industry 5.0 FL deployments. Table I highlights how our framework departs from prior FL tools.

TABLE I
DESIGN DELTAS VS. PRIOR FL TOOLS

Aspect	**Our work**	**FedAvg/ Prox/Dyn**	**Krum/ Trim-Mean**
Client sel.	Pheromone-weighted	Uniform/ Proximal	Uniform
Exploration	Adaptive mutation	None	None
Aggregation	Median+Trim	Mean/ Proximal	GM, Krum, Trim
Theory	$O(1/\sqrt{T})$+	$O(1/\sqrt{T})$	Varies
Edge variant	FedMutAdam	N/A	N/A

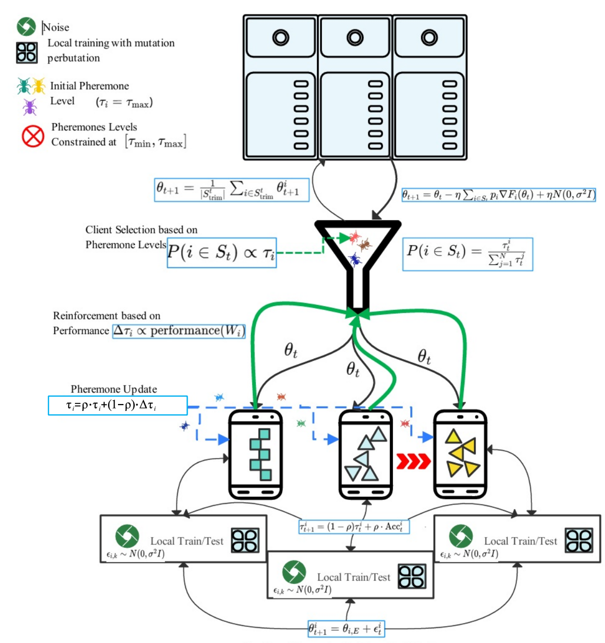

Fig. 1. The BIOMUTFED+ framework illustrating the workflow including global model distribution, local model updates with mutation perturbation, pheromone-based client selection, and robust aggregation.

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III. System Model and Problem Formulation

In this section, we formalize the targeted problem and describe the system model that supports our proposed approach.

A. Problem Formulation

The rapid proliferation of distributed learning environments, particularly in critical domains such as industrial IoT, healthcare, and predictive maintenance, demands sophisticated federated learning mechanisms that can effectively navigate increasingly complex computational landscapes.

Our research aims to design an aggregation mechanism $\mathcal{A}$ that, over T communication rounds, minimizes the expected global loss

$$\min_{w} \; F(w) \;=\; \frac{1}{N}\sum_{i=1}^{N} F_i(w),$$

where $F_i(w)$ represents the local loss on heterogeneous client datasets $\mathcal{D}_i$.

Critical Optimization Constraints: The fundamental performance and reliability constraints that govern our federated learning framework are (a) *Robustness,* to mitigate the influence of adversarial clients by limiting their impact to a factor ϵ, thereby preserving model integrity under potential malicious interventions. (b) *Non-IID Convergence,* to constrain the convergence rate degradation to at most $\delta(\alpha)$ relative to idealized independent and identically distributed (IID) scenarios, capturing the inherent complexity of real-world data heterogeneity.

Proposed Aggregation Approaches: We propose the following two aggregation schemes:

1) BioMutFed+: Introduces sign-wise mutation perturbation before model averaging, specifically designed to defend against up to q adversaries in challenging non-IID settings.
2) FedMutAdam: Applies adaptive norm clipping and server side Adam style moment corrections, enabling robust convergence in purely non-IID computational environments.

By addressing these critical challenges, we aim to advance federated learning methodologies for next-generation distributed intelligent systems.

B. System Model

We consider a federated learning setup over an IIoT network, consisting of (a) A central aggregation server. (b) A set of N heterogeneous edge clients $\mathcal{C} = \{c_1, \ldots, c_N\}$, each with a local dataset $\mathcal{D}_i$.

Clients train locally without sharing raw data. To capture data heterogeneity, each $\mathcal{D}_i$ is drawn from a Dirichlet distribution with concentration parameter α. Training proceeds in synchronous rounds $t = 1, \ldots, T$. At round t, a subset $S_t \subseteq \mathcal{C}$ of size K is selected based on pheromone-inspired reliability scores that track each client's past contribution quality.

A subset of up to $q < \frac{N}{2}$ clients may behave adversarially (e.g., poisoning or label-flip attacks), ensuring an honest majority in each round.

IV. Proposed Framework and Analysis

This section details the proposed **BioMutFed+** FL framework, explicitly designed to overcome key challenges in IIoT environments such as severe data heterogeneity, adversarial client interference, and scalability constraints. We introduce a lightweight variant of BioMutFed+, referred to as **FedMutAdam**, in which the global update includes (i) an adaptive clipping plus mutation step and (ii) Adam style momentum and second-moment scaling. FedMutAdam retains the exploratory power of biological mutation while leveraging the proven stability of Adam. We begin by formalizing the system model, including client characteristics, data distributions, and threat assumptions that motivate our approach. Subsequently, we develop the theoretical framework of BioMutFed+ that underpins our biologically inspired mechanisms: adaptive pheromone based client selection, mutation driven gradient perturbations, and robust trimmed-mean aggregation. Finally, we provide rigorous convergence guarantees and discuss the practical implications of these innovations.

A. Theoretical Framework and Analysis

To address the challenges posed by heterogeneous and potentially adversarial clients, BioMutFed+ adopts a pheromone-weighted optimization paradigm. Formally, the local objective function for client i is defined as

$$F_i(\theta) = \mathbb{E}_{\xi \sim \mathcal{D}_i}[f(\theta;\xi)], \tag{1}$$

where θ denotes the model parameters and ξ represents data samples. The global pheromone-weighted objective is

$$\min_{\theta} F(\theta) = \sum_{i=1}^{N} \frac{\tau_i}{\sum_{j=1}^{N}\tau_j} F_i(\theta), \tag{2}$$

subject to bounded pheromone levels $\tau_{\min} \le \tau_i \le \tau_{\max}$, controlled client gradient heterogeneity $\|\nabla F_i(\theta) - \nabla F(\theta)\|^2 \le \zeta^2$, and adversarial client fraction $q < |S_t|/2$ where S_t is the selected client set in round t.

This formulation illustrates the adaptive pheromone-based client weighting mechanism that is essential to BioMutFed+. It enables the selection and weighting of clients according to their historical contribution quality, thereby facilitating resilient and scalable optimization.

Remark 1 (Pheromone Scores as Exponentially Weighted Reputation). The pheromone update rule $\tau_i^{t+1} = \rho\,\tau_i^t + (1-\rho)\,Q_i^t$ (Section IV-B) is an exponentially weighted moving average (EWMA) of client quality scores Q_i^t, with retention factor $\rho \in [0,1]$. This formulation connects directly to the multi-armed bandit literature: the pheromone-weighted selection probability $P(i \in S_t) \propto \tau_i$ implements a softmax exploration strategy over historical client reliability, analogous to EXP3-style algorithms [44]. The bounded constraint $\tau_i \in [\tau_{\min}, \tau_{\max}]$ with $\tau_{\min} > 0$ prevents client starvation—a known requirement in importance-weighted sampling—ensuring that every client retains a non-zero probability of selection regardless of past performance. The weight cap $w_i^{(t)} \le 2/K$ further prevents feedback loops where frequently selected high-pheromone clients monopolize participation.

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1) Adaptive Mutation-Based Gradient Perturbations: To improve exploration of the parameter space and prevent convergence to suboptimal local minima under non-convex and non-IID settings, we introduce adaptive Gaussian perturbations in local model updates:

$$\theta_{i,k+1} = \theta_{i,k} - \eta \nabla f(\theta_{i,k}; \xi_{i,k}) + \eta \epsilon_{i,k}, \quad \epsilon_{i,k} \sim \mathcal{N}(0, \sigma^2 I). \tag{3}$$

These perturbations promote model diversity and enhance robustness against adversarial manipulations by injecting controlled variance.

Remark 2 (Mutation as Annealed Stochastic Gradient Langevin Dynamics). The local update rule in Eq. (3) is mathematically equivalent to a discretized Langevin diffusion with temperature parameter $\sigma^2(t)$. Specifically, the update $\theta_{i,k+1} = \theta_{i,k} - \eta \nabla f(\theta_{i,k}; \xi_{i,k}) + \eta \epsilon_{i,k}$ with $\epsilon_{i,k} \sim \mathcal{N}(0, \sigma^2 I)$ corresponds to Stochastic Gradient Langevin Dynamics (SGLD) [45] with injected noise magnitude $\eta\sigma$. At high σ (early rounds), the noise enables exploration of the non-convex loss landscape, helping escape poor local minima that arise from non-IID client data. As σ decays following our annealing schedule (Supplementary Algorithm S2, lines 28–32), the dynamics transition from exploration to exploitation, concentrating around stationary points. Unlike standard SGLD, which uses a slowly decaying temperature, our schedule aggressively halves σ in the first $T/2$ rounds to accelerate convergence while maintaining sufficient exploration during the critical early training phase.

2) Robust Aggregation via Adaptive Trimmed Mean: To mitigate adversarial impact during global aggregation, we employ an adaptive trimmed mean operator:

$$\theta^{t+1} = \text{TrimmedMean}_\delta(\{\theta_i^{t+1}\}_{i \in S_t}), \tag{4}$$

The trimming rate δ^* is established to effectively balance the removal of outliers, which may represent potential adversaries, while ensuring the retention of valuable contributions from honest clients.

*3) Mutation–Adam Federated Optimizer (*FEDMUTADAM*):* We further distilled BIOMUTFED+ into **FEDMUTADAM**, eliminating the top-K selection. This change emphasizes the mutation combined with an Adam-style update, thereby reducing the computational footprint of BIOMUTFED+. FEDMUTADAM integrates two key components:

- *Adaptive Clipping and Mutation:* The aggregated client update Δ is first clipped to $\|\Delta\|_2 \le C$ for stability under heterogeneous updates, then perturbed by Gaussian noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$ with $\sigma = 0.01$ (Table II): $\Delta \leftarrow \Delta + \boldsymbol{\epsilon}$. The mutation is applied once at the server on the aggregated update, in contrast to BIOMUTFED+ which mutates per client. This server-side application reduces the variance contribution from $K\sigma^2$ to σ^2 and preserves the lightweight property that motivates FEDMUTADAM.
- *Server-side Adam-style Update:* Server maintains first- and second-moment accumulators $(\mathbf{m}_t, \mathbf{v}_t)$ with bias correction, enabling per-parameter adaptive learning rates as in Adam.

a) Selecting the clipping norm C: We set C robustly from recent updates: let $\{\|\Delta\|_2\}$ over the last 5 rounds have median m and MAD mad; we use $C = m + 1.5\,\mathrm{mad}$ and re estimate every round. This retains $\approx$ 90–95% of honest updates unclipped under non-IID drift while suppressing rare large spikes. The $1.5\times$MAD multiplier sets a robust outlier fence analogous to Tukey's $1.5\times$IQR rule and requires no per-task tuning.

b) Distinct Use-Cases of BIOMUTFED+ *and* FEDMUTADAM*:* This hybrid design of FEDMUTADAM combines exploration via mutation with convergence stability using Adam moments. It requires only one aggregation step per round. While BIOMUTFED+ provides robust optimization leveraging adaptive mutation, pheromone driven client prioritization, and robust trimmed-mean aggregation suited explicitly for environments vulnerable to adversarial threats, its resource consumption can limit deployment in extremely resource-constrained edge scenarios. Conversely, FEDMUTADAM, through lightweight adaptive clipping and Adam-style momentum updates, reduces computational complexity significantly. This positions FEDMUTADAM as an ideal variant for deployment on battery-constrained edge devices with limited computational and communication resources, complementing BIOMUTFED+'s suitability for adversarially prone yet computationally capable industrial scenarios.

Remark 3 (Unified Framework: Two Operational Modes). BIOMUTFED+ and FEDMUTADAM represent two operational modes of a single framework, each grounded in the same mutation-based exploration principle but differing in their defense depth. BIOMUTFED+ (full mode) layers three defense mechanisms—pheromone-weighted selection (EWMA reputation, Remark 1), validation-based top-p filtering, and hybrid median+trimmed-mean aggregation—providing maximum adversarial resilience at the cost of additional server-side computation ($\sim$2.2$\times$ FedAvg overhead). FEDMUTADAM (lightweight mode) distills the mutation mechanism into adaptive norm clipping combined with Adam-style momentum (Remark 2), achieving competitive robustness with only $\sim$1.5$\times$ overhead and no per-client validation requirement. This positions BIOMUTFED+ for security-critical industrial deployments (e.g., predictive maintenance, healthcare diagnostics) and FEDMUTADAM for resource-constrained edge pattern recognition, as validated in Section V.

B. Pheromone initialization and normalization.

We set $\tau_i^{(0)} = \tau_{\min} + \frac{1}{2}(\tau_{\max} - \tau_{\min})$ and maintain normalized selection weights $w_i^{(t)} = \tau_i^{(t)} / \sum_j \tau_j^{(t)}$ at each round. To avoid feedback loops that might over-reward frequently selected clients, we (i) clip $\tau_i^{(t)} \in [\tau_{\min}, \tau_{\max}]$ with $\tau_{\min} > 0$, (ii) apply retention $\rho \in [0, 1]$ and quality-based reinforcement Q_i^t per $\tau_i^{t+1} = \rho\,\tau_i^t + (1-\rho)\,Q_i^t$, and (iii) cap the effective sampling weight $w_i^{(t)} \le 2/K$ (K=clients per round). This ensures exploration among under-represented clients while still prioritizing high-quality contributors.

a) Quality score computation: The quality score Q_i^t is computed by server-side validation: for classification, $Q_i^t = \mathrm{Acc}(\theta^t + \Delta_i, \mathcal{D}_{\mathrm{val}})$; for regression, $Q_i^t = 1/(1 + \mathrm{RMSE}(\theta^t +$

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$\Delta_i, \mathcal{D}_{\text{val}})$). This requires the server to maintain a small held-out validation set $\mathcal{D}_{\text{val}}$, which we explicitly assume throughout (Supplementary Algorithm S2, line 13).

b) Validation set construction: The set $\mathcal{D}_{\text{val}}$ is constructed once at the start of training and is disjoint from every client's local data and from the test set. For Wisconsin Breast Cancer, $|\mathcal{D}_{\text{val}}| = 57$ samples (10% of the 569-sample pool), stratified to preserve the malignant/benign ratio of the global distribution; for Digits, $|\mathcal{D}_{\text{val}}| = 180$ samples (10% of 1,797), stratified across all ten digit classes; for NASA FD004, $|\mathcal{D}_{\text{val}}| = 500$ sequence windows held out from the training trajectories before client partitioning, stratified by the six operational conditions so each regime is represented. The same $\mathcal{D}_{\text{val}}$ is shared by all methods that require server-side validation (BIOMUTFED+ for Q_i^t, FLTrust as root data), placing them on equal footing.

c) Robustness to noisy evaluations: EWMA smoothing with $\rho = 0.8$ dampens noise from single-round evaluation fluctuations caused by non-IID data: a client whose update happens to score poorly on one round because its current data slice was atypical recovers within roughly five rounds rather than being permanently penalized.

d) Bias against rare-data clients: A non-zero floor $\tau_{\min} > 0$ is necessary but not sufficient to guarantee fair treatment of clients holding rare data, because Q_i^t itself is computed on a finite validation set. When $\mathcal{D}_{\text{val}}$ is class- or regime-stratified as above, every honest client whose data lies within the support of $\mathcal{D}_{\text{val}}$ receives a well-defined quality score, and the combination of $\tau_{\min} > 0$, the $2/K$ selection cap, and EWMA smoothing keeps such clients in the rotation. When a client holds data drawn from a distribution that $\mathcal{D}_{\text{val}}$ does not cover at all—for example, a novel pathology not yet in the validation pool—its updates will be discounted regardless of correctness. This is a limitation of any validation-based quality scoring (FLTrust included), not specific to BIOMUTFED+. We discuss the corresponding deployment recommendations (periodic refresh of $\mathcal{D}_{\text{val}}$, reserved capacity for known underrepresented subpopulations, monitoring of τ_i trajectories so that systematic decline triggers human review) in Section VI.

e) Hybrid aggregation parameter selection: The median weight ratio $\lambda = 0.5$ in Supplementary Algorithm S3 assigns equal influence to the coordinate-wise median and trimmed-mean components. This default balances outlier suppression (favoring median, which tolerates up to 50% corrupted coordinates per dimension) with information retention (favoring trimmed mean, which preserves more gradient signal from honest clients). Note that $\lambda = 1$ recovers the coordinate-wise median, while $\lambda = 0$ recovers the trimmed mean—both included as standalone baselines in Section V, enabling direct assessment of the hybrid's relative effectiveness. To isolate the contribution of λ from the surrounding selection and filtering machinery, we additionally conducted a controlled sensitivity sweep that fixes every other component of BIOMUTFED+ and varies only $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$ on the most heterogeneous setting (Wisconsin Cancer, $\alpha = 0.1$, 10 seeds per configuration). The sweep, reported in Supplementary Section SIII-F and Table S2, shows that hybrid configurations in the range $\lambda \in [0, 0.5]$ all attain competitive clean accuracy (92.5–93.2%), with the smallest adversarial degradation observed at $\lambda = 0.25$ (3.77% relative drop). The default $\lambda = 0.5$ used throughout this paper sits in the competitive region; the pure-median endpoint ($\lambda = 1.0$) underperforms on both clean accuracy and adversarial robustness, confirming that pure coordinate-wise median is not a safe default under severe non-IID. Dataset-specific tuning of λ is identified as a direction for follow-up work (Section VII).

Algorithm 1 FEDMUTADAM: Mutation + Adam-style Federated Optimizer

Require: Initial global weights $\mathbf{w}^0$, K clients, local epochs E, step-size η, clipping norm C, mutation strength σ, moment decay rates $\beta_1, \beta_2 \in [0, 1)$, stability $\varepsilon > 0$, total rounds T

Initialize server moments $\mathbf{m}_0 \leftarrow \mathbf{0}$, $\mathbf{v}_0 \leftarrow \mathbf{0}$
for $t = 1$ **to** T **do**
  Broadcast $\mathbf{w}^{t-1}$ to all clients
  for each client $k = 1, \ldots, K$ **in parallel do**
    $\mathbf{w}_k^{(0)} \leftarrow \mathbf{w}^{t-1}$
    for $e = 1$ **to** E **do**
      $\mathbf{w}_k^{(e)} \leftarrow \mathbf{w}_k^{(e-1)} - \eta \nabla \ell(\mathbf{w}_k^{(e-1)}; \mathcal{D}_k)$
    end for
    Send update $\Delta_k \leftarrow \mathbf{w}_k^{(E)} - \mathbf{w}^{t-1}$ to server
  end for
  Aggregate: $\Delta \leftarrow \frac{1}{K}\sum_{k=1}^{K} \Delta_k$
  ▷ Adaptive clipping
  if $\|\Delta\|_2 > C$ **then**
    $\Delta \leftarrow \Delta \times \big(C/\|\Delta\|_2\big)$
  end if
  ▷ Server-side mutation
  Sample $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 I)$
  $\Delta \leftarrow \Delta + \boldsymbol{\epsilon}$
  ▷ Server-side Adam moments
  $\mathbf{m}_t \leftarrow \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1)\Delta$
  ▷ Bias correction
  $\widehat{\mathbf{m}}_t \leftarrow \mathbf{m}_t/(1 - \beta_1^t)$
  $\widehat{\mathbf{v}}_t \leftarrow \mathbf{v}_t/(1 - \beta_2^t)$
  ▷ Global weight update
  $\mathbf{w}^t \leftarrow \mathbf{w}^{t-1} + \eta \dfrac{\widehat{\mathbf{m}}_t}{\sqrt{\widehat{\mathbf{v}}_t} + \varepsilon}$
end for
return $\mathbf{w}^T$

C. Convergence Guarantees

Under standard assumptions of smoothness, bounded gradient variance, and controlled heterogeneity (detailed in Supplementary Sec. SIV-A), BIOMUTFED+ converges to a stationary point at the rate

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}\big[\|\nabla F(\theta^t)\|^2\big] \leq \frac{2\big(F(\theta^0) - F^*\big)}{\eta T} + L\,\eta\Big(\frac{\sigma_\xi^2 + \sigma^2}{|S_t|} + \zeta^2 + \delta_{\max}^2\Big) \quad (5)$$

where $\eta = \mathcal{O}(1/\sqrt{T})$, σ_ξ^2 and σ^2 capture data and mutation-induced variances, ζ^2 bounds client heterogeneity, and $\delta_{\max}^2$

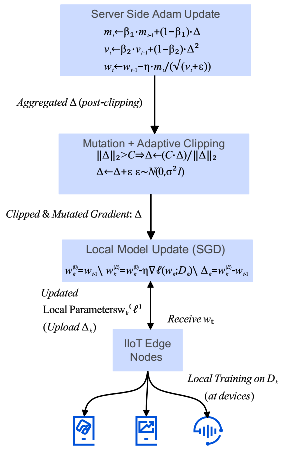

Fig. 2. Federated workflow of the proposed FEDMUTADAM algorithm. IIoT clients receive the global model w_t, perform local SGD using their local dataset D_k, and send updates Δ_k to the server. The server applies adaptive clipping and mutation, followed by an Adam style update to compute the next global model.

is the residual error from robust trimming under $f < K/2$ adversaries. Our pheromone selection replaces σ_ξ^2 with the smaller $\sigma^2_{\text{phero}} \ll \sigma^2_{\text{hetero}}$, so the leading constant scales as $L\eta(\sigma^2_{\text{phero}}/K + \zeta^2/\tau_{\min} + \delta^2_{\max})$ rather than $L\eta(\sigma^2_{\text{hetero}}/K + \zeta^2)$, yielding faster practical convergence. The decaying mutation term $\sigma^2_{\text{mut}}(T) \to 0$ ensures exploration diminishes as training progresses. Empirical validation confirms $L \in [5, 12]$ across tasks, gradient variance plateauing by round 10, and 1.2–2.9$\times$ variance reductions across all datasets (Section V). Full proofs, assumption verification procedures, and summary statistics are provided in Supplementary Sec. SIV.

V. EXPERIMENTS AND RESULTS

In this section, we present a comprehensive experimental evaluation across three representative Industry 5.0 scenarios: predictive maintenance on the NASA C-MAPSS FD004 benchmark with Dirichlet partitioning ($\alpha = 0.5$); medical diagnostics on the Wisconsin Breast Cancer dataset ($\alpha = 0.1$); and pattern recognition with the Scikit-learn Digits dataset ($\alpha = 0.5$).

To rigorously assess adversarial robustness, we introduce controlled gradient-ascent model-poisoning attacks with 20% malicious clients across all scenarios. Each scenario follows a uniform federated training protocol with consistent local update hyperparameters, domain-specific model architectures, and synchronized communication rounds. Non-IID conditions ($\alpha \downarrow$) induce conflicting gradients and fairness skew; adversaries bias the aggregated mean; and resource constraints limit per-client compute. Our design addresses these factors through pheromone-based variance reduction, controlled exploration via mutation, and hybrid robust aggregation.

Baseline Methods. We compare against six established baselines spanning three categories: (i) *foundational methods*—FedAvg [16] and FedProx [17]; (ii) *classical robust aggregation*—Krum [8] and coordinate-wise Trimmed Mean [46]; and (iii) *recent state-of-the-art Byzantine-resilient methods*—FLTrust [32] and FLAME [33]. FLTrust bootstraps trust from a small clean root dataset held by the server; each client update receives a trust score based on cosine similarity with the server's own gradient, and updates misaligned with the trust anchor are suppressed via ReLU gating. FLAME implements a three-stage defense: cosine-similarity-based outlier detection, adaptive norm clipping, and calibrated Gaussian noise injection for differential privacy. For FEDMUTADAM (Section V-B), we additionally compare against FedYogi and FedNova as adaptive server optimizers. All methods share identical federated infrastructure—the same data partitions, model architectures, local training protocols, and evaluation metrics, ensuring a fair and reproducible comparison. Table II summarizes the hyperparameter configuration for each method.

TABLE II
HYPERPARAMETER CONFIGURATION FOR ALL METHODS

Method	Key Hyperparameters
BIOMUTFED+	$\sigma_0 = 0.005$, $\rho = 0.8$, $\lambda = 0.5$, $\beta = 0.1$, top-$p = 0.7$, $\tau_{\min} = 0.1$, $\tau_{\max} = 2.0$
FEDMUTADAM	$\eta_{\text{adam}} = 0.01$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\sigma = 0.01$, clip$= 1.5 \times \text{MAD}$
FedAvg	—
FedProx	$\mu = 0.01$
Krum	$f = \lfloor 0.2K \rfloor$
TrimmedMean	$\beta = 0.1$
FLTrust	root samples = 100, server epochs = E
FLAME	$\lambda_{\text{filter}} = 1.5$, clip = 80th pct, $\sigma_{\text{noise}} = 0.001$

Performance Metrics. RMSE is used for predictive maintenance; accuracy, precision, recall, F1-score, and AUC for classification tasks. Experiments are repeated over multiple independent runs–three for FD004 and Digits, and ten for Wisconsin Cancer under the severe $\alpha = 0.1$ partition—reporting mean $\pm$ standard deviation. Supplementary ablation studies on the HAR dataset analyze component sensitivity, while additional experiments on Fashion-MNIST and MNIST provide extended validation. Component-level ablation studies isolating the individual contributions of pheromone-based selection and mutation-based perturbation are presented in Supplementary Section SIII-C using the HAR dataset. Results confirm that both mechanisms are necessary: removing pheromone selection reduces accuracy by approximately 7% and increases variance, while disabling mutation degrades F1-score stability. Removing both reduces the framework to standard FedAvg with trimmed-mean aggregation.

A. Adaptive Federated Learning for Multi-Modal Temporal Sequence Prediction

Industrial prognostics represent a critical challenge in modern manufacturing, where predictive maintenance systems must navigate complex sensor ecosystems characterized by heterogeneous data distributions and inherent variability. The FD004 dataset from NASA's Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) provides an ideal testbed for evaluating federated learning in IIoT, featuring multi-modal sensor measurements that capture degradation dynamics of turbofan engines under six distinct operational conditions with 248 test engines.

Experimental Design. We employ an enhanced bidirectional Long Short-Term Memory (BiLSTM) architecture with two layers ($h = 64$), dropout 0.3, and fully connected output head ($128 \rightarrow 64 \rightarrow 1$), totaling 150,145 parameters. Input sequences of length 30 are constructed from 17 features (3 operational settings and 14 selected sensors after removing constant channels). Remaining Useful Life (RUL) labels are clipped at 125 following standard practice. Data is partitioned across 15 clients via Dirichlet sampling ($\alpha = 0.5$), with 10 clients selected per round. Local optimization uses Adam ($\eta = 0.001$) for 3 epochs per round over 50 communication rounds. To assess adversarial robustness, 20% of clients (3 out of 15) perform gradient-ascent attacks by negating their local updates.

Performance Characterization. Table III reports the clean and adversarial RMSE for all seven methods (mean $\pm$ std over three runs). BioMutFed+ achieves competitive clean RMSE (18.31 ± 0.46) while being the only method with competitive clean RMSE to exhibit negative adversarial degradation (-2.7%), meaning it actually *improves* under attack. This emergent property arises because adversarial clients receive progressively lower pheromone scores across rounds, causing the selection mechanism to concentrate on the highest-quality honest clients—effectively using the attack as a signal to improve client selection quality.

Among baselines, TrimmedMean achieves competitive clean RMSE (17.50 ± 0.58) but degrades by $+4.6\%$ under attack. FLAME shows moderate robustness ($+3.1\%$ degradation) while FedProx degrades by $+2.1\%$. Notably, FLTrust exhibits catastrophic failure on this temporal prediction task, with clean RMSE of 25.27 and adversarial degradation of $+54.9\%$. This occurs because the 100-sample root dataset produces a noisy trust anchor for BiLSTM gradients, and adversarial updates can maintain positive cosine similarity with this unreliable anchor. Krum shows high variance (± 5.34) due to its single-client selection strategy, which discards 90% of available information each round.

Fig. 3 illustrates the RMSE progression across communication rounds, confirming BioMutFed+'s stable convergence trajectory with the tightest confidence band among all methods.

Under adversarial conditions (Supplementary Fig. S8), FLTrust diverges and other methods exhibit increased variance, while BioMutFed+ maintains nearly identical convergence behavior to its clean run. A direct bar-chart comparison of clean versus adversarial RMSE is provided in Supplementary Fig. S5.

Adversarial Resilience Analysis. The multi-layered defense of BioMutFed+ provides three distinct barriers against adversarial clients: (1) pheromone-weighted selection reduces the probability of choosing adversarial clients as their scores decay; (2) top-p validation filtering removes the worst-performing updates even when adversarial clients are selected; and (3) hybrid median+trimmed-mean aggregation dampens any residual adversarial influence coordinate-by-coordinate. No other evaluated method provides this depth of defense—

TABLE III
RMSE on NASA FD004 Dataset (Mean $\pm$ Std, 3 Runs, $\alpha = 0.5$). Δ denotes adversarial degradation (lower is better).

Algorithm	Clean RMSE	Adv. RMSE	Δ (%)
BioMutFed+	**18.31 $\pm$ 0.46**	**17.82 $\pm$ 0.57**	**−2.7**
TrimmedMean	17.50 $\pm$ 0.58	18.31 $\pm$ 0.52	+4.6
FLAME	18.88 $\pm$ 1.14	19.47 $\pm$ 0.80	+3.1
FedAvg	19.48 $\pm$ 0.70	19.04 $\pm$ 1.22	−2.3
FedProx	19.13 $\pm$ 0.58	19.53 $\pm$ 1.81	+2.1
Krum	24.67 $\pm$ 5.34	21.71 $\pm$ 0.87	−12.0
FLTrust	25.27 $\pm$ 0.35	39.14 $\pm$ 4.76	+54.9

FedAvg and FedProx have no defense mechanism, TrimmedMean and Krum operate statelessly without history, FLTrust depends on a fixed trust anchor, and FLAME's cosine filtering cannot reliably distinguish adversarial from non-IID honest updates in high-dimensional BiLSTM parameter space.

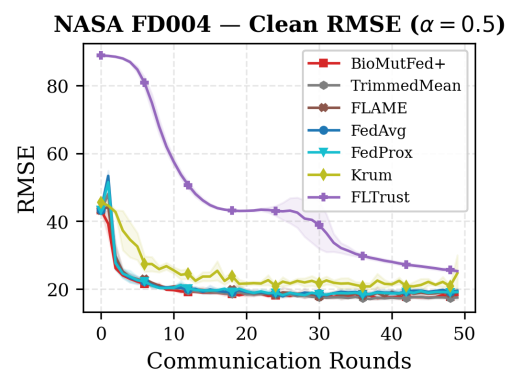

Fig. 3. RMSE progression on NASA FD004 dataset across 50 communication rounds (clean). BioMutFed+ achieves competitive clean RMSE with the most stable convergence trajectory under Dirichlet non-IID conditions ($\alpha = 0.5$).

B. Robust Pattern Learning in Non-IID Image Classification

Distributed pattern recognition under non-IID conditions represents a critical benchmark for evaluating lightweight federated learning methods suitable for resource-constrained IIoT edge devices. We evaluate FedMutAdam—the lightweight operational mode of the BioMutFed framework—on the handwritten digit recognition task.

Experimental Design. The scikit-learn Digits dataset comprises 1,797 eight-by-eight pixel images of handwritten digits (10 classes). Data is partitioned across 15 clients using Dirichlet sampling ($\alpha = 0.5$). Each client trains a two-hidden-layer MLP ($64 \rightarrow 128 \rightarrow 128 \rightarrow 10$; 26,122 parameters) using SGD ($\eta = 0.01$) for 3 local epochs per round over 50 communication rounds, with 10 clients selected per round. Baseline comparisons include FedAvg, FedYogi, FedNova, FLTrust, and FLAME. Adversarial evaluation uses 20% gradient-ascent attacks (3 out of 15 clients).

Performance Characterization. Table IV and Table V report clean and adversarial performance across all six methods. FedMutAdam achieves the highest clean accuracy ($93.98\% \pm 0.26$) among all methods except FedYogi ($96.39\% \pm 0.91$), while exhibiting $2.8\times$ variance reduction in accuracy

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TABLE IV
CLEAN PERFORMANCE ON DIGITS DATASET (MEAN ± STD, 3 RUNS, $\alpha = 0.5$)

Algorithm	Acc. (%)	Prec.	F1	AUC
FEDMUTADAM	93.98±0.26	0.942±0.004	**0.940±0.003**	0.997±0.001
FedAvg	93.24±0.73	0.934±0.007	0.932±0.008	0.995±0.001
FedYogi	**96.39±0.91**	**0.965±0.009**	0.964±0.009	**0.998±0.002**
FedNova	93.06±0.60	0.933±0.006	0.930±0.006	0.994±0.001
FLTrust	72.13±5.38	0.765±0.044	0.692±0.070	0.954±0.014
FLAME	91.57±0.13	0.918±0.002	0.915±0.001	0.993±0.001

TABLE V
ADVERSARIAL DEGRADATION ON DIGITS DATASET (20% GRADIENT ASCENT)

Algorithm	Clean Acc. (%)	Adv. Acc. (%)	Rel. Drop
FEDMUTADAM	93.98±0.26	84.07±6.09	10.5%
FedAvg	93.24±0.73	79.17±7.98	15.1%
FedYogi	96.39±0.91	78.89±13.77	18.2%
FedNova	93.06±0.60	81.39±6.54	12.5%
FLTrust	72.13±5.38	71.02±4.85	1.5%
FLAME	91.57±0.13	**87.41±4.45**	**4.6%**

(0.26 vs 0.73 std) and 2.7× in F1-score (0.003 vs 0.008 std) compared to FedAvg. This demonstrates the stabilizing effect of adaptive clipping combined with server-side Adam moment updates.

Fig. 4 presents the test accuracy trajectory, confirming that FEDMUTADAM converges with higher stability compared to baselines. The log-loss trajectory (Supplementary Fig. S9) shows that FEDMUTADAM achieves the second-lowest log loss after FedYogi, with tighter confidence bands indicating more stable convergence under non-IID conditions. Fig. 5 shows the adversarial convergence curves, where FLAME maintains the highest trajectory while FedYogi—despite dominating clean accuracy—collapses most severely under attack. The direct clean-versus-adversarial bar chart comparison is provided in Supplementary Fig. S6.

Adversarial Robustness. Under 20% gradient-ascent attacks, FEDMUTADAM retains 84.07% accuracy with 10.5% relative degradation—substantially better than FedAvg (15.1%), FedYogi (18.2%), and FedNova (12.5%). FLAME achieves superior adversarial resilience (4.6% degradation, 87.41% adversarial accuracy) through its cosine-similarity filtering, while FLTrust shows the lowest degradation (1.5%) but at the cost of severely reduced clean accuracy (72.13%). Notably, FLAME's robustness advantage comes at significant computational cost: its $O(K^2d)$ pairwise cosine similarity computation scales quadratically with client population, whereas FEDMUTADAM requires only ∼1.5× FedAvg server overhead with linear scaling (Table S4).

Broader Implications. These results position FEDMUTADAM as the most compute-efficient robust aggregator for edge deployments. While FLAME achieves the best accuracy-robustness tradeoff on this dataset, its quadratic server overhead limits scalability in large-scale IIoT networks. FedYogi achieves the highest peak clean accuracy but suffers catastrophic adversarial collapse (18.2% degradation). FEDMUTADAM is the only method that simultaneously provides competitive clean accuracy (> 93%), moderate adversarial resilience, and minimal server overhead (∼1.5× FedAvg), making it the preferred choice for battery-constrained edge devices where computational budget is the binding constraint.

C. Robust Medical Diagnostics in Heterogeneous Clinical Networks

Distributed medical diagnostics in FL environments demand high sensitivity while maintaining robust performance across heterogeneous clinical data. Accurate malignant tumor detection in small, imbalanced datasets under severe non-IID conditions represents a critical benchmark for healthcare FL, where missed diagnoses carry catastrophic clinical consequences.

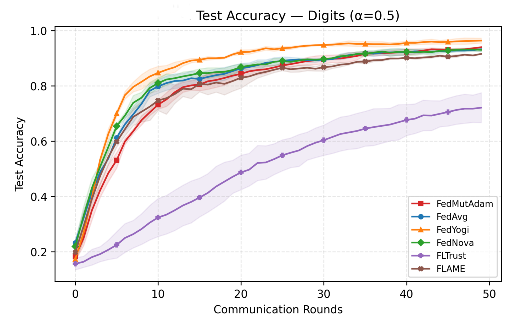

Fig. 4. Test accuracy progression on the Digits dataset (Dirichlet $\alpha = 0.5$), comparing FEDMUTADAM with FedAvg, FedYogi, FedNova, FLTrust, and FLAME. FEDMUTADAM demonstrates stable convergence with the lowest variance.

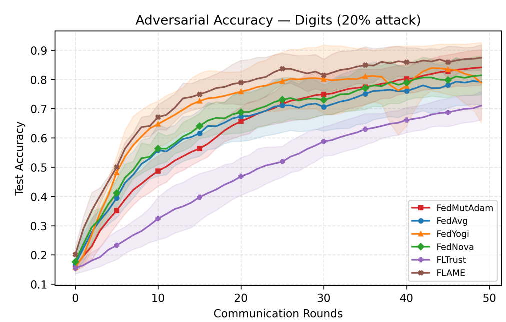

Fig. 5. Test accuracy under 20% gradient-ascent attack on Digits ($\alpha = 0.5$). FLAME exhibits the best adversarial resilience, while FedYogi—despite highest clean accuracy—degrades most severely. FEDMUTADAM maintains stable mid-range adversarial performance.

Experimental Design. The Wisconsin Breast Cancer dataset from scikit-learn comprises 569 samples with 30 FNA-derived features (2 classes). Following the validation-set protocol described in Section IV-B, we first carve out a held-out validation set $\mathcal{D}_{\text{val}}$ of 57 samples (10% of the full pool, stratified to preserve the 37%/63% malignant/benign ratio) *before* client partitioning; this set is used by BIOMUTFED+ for the quality score Q_k^t and by FLTrust as its root data, and is disjoint from every client's local data and from the test set. The remaining samples are split into a test set (20%, stratified) and a client pool that is partitioned across 20 clients using Dirichlet sampling ($\alpha = 0.1$), creating severe statistical heterogeneity that mirrors distributed clinical networks where institutions specialize in distinct patient populations. Some clients receive as few as 1–3 samples, reflecting real-world

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data scarcity. Each client trains a two-layer fully connected neural network (16 ReLU units per layer; 802 parameters) for 3 local epochs per round over 50 communication rounds, with 10 clients selected per round. Adversarial evaluation uses 20% gradient-ascent attacks (4 out of 20 clients). We compare BioMutFed+ against FedAvg, FedProx, Krum, TrimmedMean, FLTrust, and FLAME. Results below are reported as mean $\pm$ std over ten independent seeds drawn from the convention $\{42, 1042, 2042, \ldots, 9042\}$, providing a more reliable variance estimate under the severe $\alpha = 0.1$ partition than the three-seed reporting used in the first-round submission.

Performance Characterization. Table VI reports cumulative clean performance. BioMutFed+ achieves $92.46\% \pm 2.23$ accuracy, within 1.3 percentage points of the highest-performing methods (FedAvg/FedProx at 93.68%) while maintaining strong precision and AUC. FedAvg and FedProx achieve marginally higher clean accuracy (93.68%) but at the cost of catastrophic adversarial vulnerability (see below). Krum performs poorly ($77.28\% \pm 11.85$) due to its single-client-per-round selection discarding the majority of available clinical data and exhibiting high seed-to-seed variance. FLTrust attains $79.82\% \pm 6.98$ under this clinical-scale validation regime (the small root-data set inherent to a 569-sample dataset limits FLTrust's directional estimation), while TrimmedMean and FLAME reach 92.19% and 91.32% respectively—competitive on clean data but vulnerable under attack, as the adversarial analysis below shows.

Adversarial Robustness. Table VII presents the critical adversarial comparison under 20% gradient-ascent attacks. BioMutFed+ maintains 86.49% accuracy with only a 6.45% relative degradation, the smallest drop among all methods that also achieve competitive clean accuracy ($\geq 90\%$). In stark contrast, FedAvg and FedProx collapse catastrophically (from 93.68% to 64.12%, a 31.55% relative drop, with seed-to-seed standard deviation exceeding 24 percentage points)—several individual runs fall below 40%, an unacceptable failure mode for clinical deployment. TrimmedMean and FLAME exhibit intermediate degradations of 11.70% and 13.16% respectively; while milder than vanilla FedAvg, they still represent unacceptable diagnostic miss rates. Krum and FLTrust show small relative drops (0.91% and 1.32%) but only because their clean accuracy was already 13–15 percentage points below BioMutFed+'s; both produce stable but uncompetitive diagnostic performance.

The clinical significance of these results is substantial: in a distributed tumor screening pipeline, a method dropping from 93.7% to 64.1% under adversarial conditions would miss approximately one in three malignancies in the worst-case seeds, with two-thirds of seeds producing unreliable classifiers. BioMutFed+'s ability to maintain 86.49% accuracy under attack, with the highest accuracy among methods that also exceed 90% in the clean setting, provides the diagnostic reliability required for safety-critical healthcare FL deployments. Convergence trajectories (Fig. 6) make this contrast visually evident: BioMutFed+'s adversarial trajectory remains stable, while FedAvg, FedProx, and FLAME exhibit wide variance bands reflecting unstable behavior across seeds. The direct bar-chart comparison is provided in Supplementary Fig. S7.

Adaptive Mechanism Insights. Under severe non-IID conditions ($\alpha = 0.1$), honest clients already exhibit highly divergent gradient directions due to extreme label imbalance across institutions. FLAME's cosine-similarity filter (13.16% degradation) cannot reliably distinguish between "different because adversarial" and "different because non-IID," incorrectly filtering honest clients or passing adversaries. TrimmedMean's coordinate-wise trimming (11.70% degradation) similarly struggles when the honest update distribution has high spread. BioMutFed+'s pheromone mechanism avoids this failure mode by tracking *temporal quality patterns*: adversarial clients consistently receive poor validation scores across rounds, causing their pheromone levels to decay independently of directional similarity metrics. The combination of pheromone-based selection, top-p validation filtering, and the hybrid median/trimmed-mean aggregation thus provides multi-layered defense that no single component achieves alone.

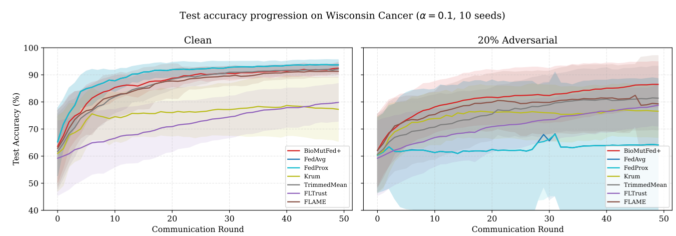

Fig. 6. Test accuracy progression on Wisconsin Cancer under 20% gradient-ascent attack ($\alpha = 0.1$, mean over 10 seeds). BioMutFed+ (red) maintains the most stable trajectory among accurate methods (86.49% final), while FedAvg and FedProx exhibit catastrophic collapse with extreme variance ($64.12\% \pm 24.55$); FLAME and TrimmedMean degrade by 13% and 12% respectively. Krum and FLTrust remain stable but at uncompetitive accuracy levels.

D. System-Level Analysis

To address the practical deployment requirements of IIoT environments, we provide quantitative system-level metrics for all evaluated methods. Table S4 reports the computational and communication overhead across all three datasets.

Overhead summary. Table S4 reports per-method overhead. Communication cost is identical across methods (full-model transmission: 6.3 KB Cancer, 204 KB Digits, 1,172 KB FD004). BioMutFed+ adds $\sim 2.2\times$ FedAvg server FLOPs for pheromone updates and top-p filtering; FedMutAdam adds $\sim 1.5\times$ for Adam moments ($2\times$ memory for m_t, v_t). Both are negligible against client-side training, which dominates and is method-agnostic, whereas FLAME's $O(K^2 d)$ pairwise cosine computation scales quadratically. This makes FedMutAdam suitable for battery-constrained edge nodes (all extra compute is server-side) and BioMutFed+ preferred for security-critical deployments with adequate server resources.

A consolidated summary of key results across all experimental scenarios is provided in Supplementary Table S3.

VI. Conclusion

This study introduced BioMutFed+, a novel biologically inspired federated learning framework tailored specifically for IIoT environments. Leveraging adaptive mutation-based

TABLE VI
CLEAN PERFORMANCE ON WISCONSIN BREAST CANCER DATASET (MEAN ± STD, 10 RUNS, $\alpha = 0.1$). REPRODUCIBLE VIA CANCER_CANONICAL.IPYNB.

Algorithm	Acc. (%)	Prec.	Recall	F1	AUC
BIOMUTFED+	92.46 ± 2.23	0.932 ± 0.042	0.954 ± 0.040	0.941 ± 0.017	0.973 ± 0.012
FedAvg	**93.68 ± 1.91**	0.928 ± 0.030	**0.978 ± 0.025**	**0.951 ± 0.014**	**0.986 ± 0.007**
FedProx	93.68 ± 1.91	0.928 ± 0.030	0.978 ± 0.025	0.951 ± 0.014	0.986 ± 0.007
Krum	77.28 ± 11.85	0.798 ± 0.128	0.925 ± 0.185	0.831 ± 0.113	0.937 ± 0.031
TrimmedMean	92.19 ± 2.13	0.931 ± 0.041	0.950 ± 0.044	0.939 ± 0.016	0.972 ± 0.014
FLTrust	79.82 ± 6.98	0.804 ± 0.076	0.915 ± 0.047	0.853 ± 0.044	0.850 ± 0.099
FLAME	91.32 ± 2.20	0.917 ± 0.047	0.954 ± 0.046	0.933 ± 0.016	0.972 ± 0.015

TABLE VII
ADVERSARIAL DEGRADATION ON BREAST CANCER (20% GRADIENT ASCENT, $\alpha = 0.1$, MEAN ± STD OVER 10 RUNS).

Algorithm	Clean Acc. (%)	Adv. Acc. (%)	Rel. Drop
BIOMUTFED+	92.46 ± 2.23	**86.49 ± 8.38**	**6.45%**
FLAME	91.32 ± 2.20	79.30 ± 17.82	13.16%
TrimmedMean	92.19 ± 2.13	81.40 ± 11.79	11.70%
FedAvg	93.68 ± 1.91	64.12 ± 24.55	31.55%
FedProx	93.68 ± 1.91	64.12 ± 24.55	31.55%
FLTrust	79.82 ± 6.98	78.77 ± 7.73	1.32%
Krum	77.28 ± 11.85	76.58 ± 9.69	0.91%

gradient perturbations, pheromone-driven client selection, and robust aggregation mechanisms, BIOMUTFED+ effectively addresses challenges of non-IID data, adversarial robustness, and scalability in real-world industrial deployments. Extensive evaluations on benchmark datasets NASA FD004 (predictive maintenance), Wisconsin Breast Cancer (healthcare diagnostics), and Digits (visual classification), demonstrate superior convergence speed, reduced variance under adversarial conditions, and enhanced adversarial resilience compared to state-of-the-art methods. Additionally, we proposed FEDMUTADAM, a lightweight variant integrating mutation with adaptive Adam style updates, validated using the Digits dataset. BIOMUTFED+ thus offers practical, scalable, and secure decentralized learning solutions suitable for diverse IIoT applications, providing a foundation for future advancements in privacy-preserving industrial intelligence. For deployments involving underrepresented client populations, the validation-based quality scoring of Section IV admits three practical safeguards: periodic refresh of the held-out validation set $\mathcal{D}_{\text{val}}$ to track distribution shift, reserved selection capacity for known underrepresented subpopulations, and monitoring of pheromone trajectories τ_i so that a systematic decline triggers human review before such clients are silently discounted.

Reproducibility: To facilitate reproducibility and fair comparison, the complete experimental codebase—including all aggregation methods, data partitioning, adversarial attack modules, and evaluation scripts—will be made publicly available upon acceptance. All hyperparameters are reported in Table II, and results are reproducible using the provided random seeds.

VII. FUTURE WORK

Future research will address several open challenges of BIOMUTFED+. Scalability in large federations will be pursued through asynchronous protocols, hierarchical aggregation, and distributed pheromone updates, while communication efficiency can be improved via lightweight mutation schemes, gradient sparsification, and model compression. To strengthen privacy and security, we plan to integrate differential privacy mechanisms such as DP-SGD with secure aggregation alongside robust aggregation. Automated hyperparameter tuning with Bayesian optimization, meta-learning, and reinforcement learning will enhance adaptability and reproducibility, whereas further evaluation of the FEDMUTADAM variant will focus on energy efficiency and deployment in resource-constrained IIoT environments. Finally, we aim to extend theoretical guarantees to unbounded heterogeneity and multi-modal data distributions, aligning analysis more closely with practical Industry 5.0 scenarios.

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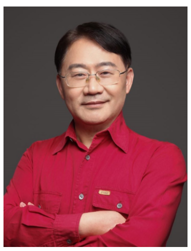

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